

1) Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Step 1: Start Wireshark

- Open Wireshark and select the active network interface, such as Wi-Fi or Ethernet.
- Start the packet capture.

Step 2: Generate Traffic

- Open a web browser and refresh the Flipkart homepage.
- Stop the capture after the page has loaded so the traffic can be examined.

Step 3: Identify the TCP/IP Layers

- Application Layer — Look for an application protocol such as HTTP, DNS, or TLS in the packet details. For a modern HTTPS website, TLS is commonly visible rather than plain HTTP.
- Transport Layer — Look for TCP or UDP. For a TCP packet, fields such as Source Port, Destination Port, Sequence Number, and Acknowledgment Number identify the Transport layer.
- Internet Layer — Look for IPv4 or IPv6 and inspect Source and Destination IP addresses.
- Network Access Layer — Look for Ethernet II or Wi-Fi/802.11 details and inspect source and destination MAC addresses.

Packet Identification Table

- Application: Example protocol — DNS or TLS/HTTP-related traffic, depending on the captured packets.
- Transport: TCP — identified by source/destination ports and TCP sequence/acknowledgment fields.
- Internet: IPv4/IPv6 — identified by source and destination IP addresses.
- Network Access: Ethernet II or Wi-Fi — identified by MAC addresses and link-layer fields.

2) Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use.

OSI and TCP/IP Comparison

- OSI Application → TCP/IP Application — Example: using WhatsApp, a browser, or Zomato involves application-level data and protocols.
- OSI Presentation → TCP/IP Application — Example: data formatting, encryption, or encoding used by an application.
- OSI Session → TCP/IP Application — Example: maintaining an application communication session.
- OSI Transport → TCP/IP Transport — Example: TCP carrying application data between endpoints.
- OSI Network → TCP/IP Internet — Example: IP addressing and routing packets toward a destination server.
- OSI Data Link → TCP/IP Network Access — Example: Ethernet or Wi-Fi frames moving data across the local network.
- OSI Physical → TCP/IP Network Access — Example: electrical, radio, or optical transmission of bits.

Side-by-Side Summary

- OSI has 7 layers: Application, Presentation, Session, Transport, Network, Data Link, Physical.
- TCP/IP commonly uses 4 layers: Application, Transport, Internet, Network Access.
- The TCP/IP Application layer combines the functions of the OSI Application, Presentation, and Session layers.
- The TCP/IP Network Access layer broadly covers functions represented by the OSI Data Link and Physical layers.

Real-World Example

- When you send a WhatsApp message, the application creates the message data, the Transport layer can use TCP or another transport mechanism depending on the service, IP handles addressing/routing, and Wi-Fi/Ethernet carries the data across the local network.

3) Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1–2 lines.

TCP Transport-Layer Fields

- Source Port — Identifies the application/service port on the sending side.
- Destination Port — Identifies the application/service port on the receiving side.
- Sequence Number — Identifies the position of TCP data in the byte stream and helps TCP maintain the correct order.
- Acknowledgment Number — Indicates the next byte the sender expects to receive when acknowledgment is valid.
- Header Length / Data Offset — Indicates the size of the TCP header.
- Flags — Control TCP functions such as connection setup, acknowledgment, and termination. Common flags include SYN, ACK, FIN, and RST.
- Window Size — Indicates how much data the receiver is currently prepared to accept, supporting flow control.
- Checksum — Helps detect errors in the TCP segment.
- Urgent Pointer — Used with the URG flag to indicate urgent data information.
- Options — Provides optional TCP features and parameters when present.

How to Label the Trainer's Screenshot

- Locate the TCP header section in the packet details.
- Mark Source Port, Destination Port, Sequence Number, Acknowledgment Number, Flags, Window Size, Checksum, and other TCP fields shown in the supplied screenshot.
- Use the exact field names displayed by the trainer's screenshot when labeling the final answer.

4) Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms.

Layer-by-Layer Process

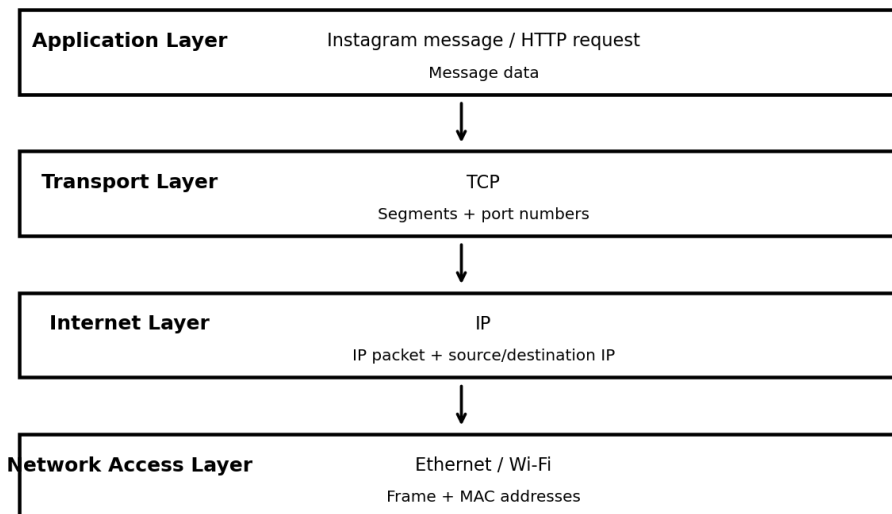
- Application Layer — You write and send the Instagram message. The application prepares the message data for transmission.
- Transport Layer — The data is prepared for end-to-end transport. With TCP, ports and sequencing information are added to form a TCP segment.
- Internet Layer — IP addressing information is added so routers can forward the packet toward the destination server.
- Network Access Layer — The packet is placed into an Ethernet or Wi-Fi frame and transmitted over the local network.

Encapsulation

- Application data → TCP segment → IP packet → Ethernet/Wi-Fi frame.
- At the receiving side, the process is reversed: the frame is processed, IP information is examined, the Transport header is processed, and the application receives the message data.

Reference Diagram

Example: Sending a Message from a Device to an Internet Server



- The diagram below illustrates this process using simple application-related terms.

5) Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app it represents.

Example Packet Analysis

- Selected packet: TCP packet captured while loading an Internet application.
- TCP/IP layer: Transport layer.
- How it was identified: The packet details show TCP fields such as Source Port, Destination Port, Sequence Number, Acknowledgment Number, and Flags.
- Real-world action: The TCP traffic can represent part of the communication used while an application loads data from an Internet service.
- Packet number, source/destination addresses, and exact protocol details should be taken from your own Wireshark capture.

Student Packet Record

- Packet Number: _____
- Protocol: _____
- TCP/IP Layer: _____
- Source IP: _____
- Destination IP: _____
- Source Port: _____
- Destination Port: _____
- How I identified the layer: _____
- Real-world app action: _____